Exact Spatial Flatness and the Spin–Torsion Link: an Open Problem in Einstein–Cartan Cosmology

Pascal Fichant[1]

[1] *Independent Researcher, Montpellier, France (p.fichant.research@gmail.com)*

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Standard cosmology treats the observed spatial flatness of the Universe ($\Omega_K \simeq 0$) as an observational coincidence, to be explained dynamically by inflation. We adopt a different stance and state it as a postulate: spatial flatness is *exact*, $\Omega_K \equiv 0$, as a topological property of spacetime rather than a dynamical outcome. Unlike inflationary flattening, which predicts only $\Omega_K \approx 0$, this is a sharp and falsifiable claim: a statistically robust detection of non-zero spatial curvature — one that survives systematics and the model degeneracies of the inference — would refute the postulate of exact flatness. *(The falsifiable content is the exactness of the postulate, not the mathematical existence of a non-zero central value at arbitrary precision; and the refutation bears on the postulate, not on those ECF sectors that do not rest on it.)*

We are equally explicit about what this postulate does *not* do. Because $\sum_i \Omega_i = 1$ is an identity once $k = 0$ — the critical density $\rho_{\rm crit} = 3H^2/8\pi G$ being recomputed with H — the closure condition cannot enforce any compensating dark-energy component, and constrains neither $\rho_{\rm DE}$ nor its equation of state. Earlier versions of this Letter presented such a compensation as a mechanism; that claim is withdrawn.

What Einstein–Cartan cosmology does supply, independently of this postulate, is a *static* dark-energy term derived from the action: the FLRW reduction of the axial torsion sector yields a background condensate with $w = -1$, its fluctuations behaving as radiation. The observed *evolution* of the equation of state does not follow, and remains calibrated.

This leaves a well-posed open problem, which we state rather than solve. The Weyssenhoff spin fluid that reduces the sound horizon before recombination, and the frozen condensate that sources dark energy, are two regimes of the *same* torsion field. A relation between $\Omega_{\rm spin}$ and Ω_τ is therefore expected on physical grounds — but it must be derived from the action, not imposed by Ω-bookkeeping. We formulate this problem precisely, show why the route through the closure condition fails, and identify the propagating torsion mode as the direction in which a derivation could be sought.

Version note (v9). *The Mahalanobis distance of the calibrated point to the DESI DR1 central value is corrected from 2.22σ to 2.03σ (full covariance, twelve observables; the earlier value came from a withdrawn thirteen-observable block). The calibrated point (w_0, w_a) is now quoted without the error bars of the likelihood profile. No other change to the text. The note of version v8 follows.*

Version note (v8). *This version withdraws the compensation mechanism presented in v4–v7 and reformulates the Letter around the open problem it leaves. The postulate is retained, with its scope restricted to exact spatial flatness. Supersedes preprint v5 (Zenodo, concept doi:10.5281/zenodo.19798923). Key changes: (i) the closure condition is shown to be an identity, incapable of enforcing a compensating debt; (ii) the static dark-energy term is attributed to the FLRW reduction of the action (Foundation II, Appendix J), independently of the postulate; (iii) the $\Omega_{\rm spin}$–Ω_τ relation is stated as the central open question rather than assumed; (iv) the analogy with charge conservation is withdrawn.*

Keywords: Einstein-Cartan cosmology; Topological Invariance Principle; dark energy; torsion; spatial flatness; cosmological constant; spin-torsion coupling; Weyssenhoff fluid; geometric gauge postulate; non-singular bounce; QCD phase transition; DESI; cosmic birefringence; Hubble tension; S_8 tension.

INTRODUCTION

Precision cosmology, largely driven by Cosmic Microwave Background (CMB) measurements from Planck+BAO [3], dictates that the observable Universe is remarkably flat, with the curvature parameter $\Omega_K = 0.0007 \pm 0.0019$. Within the standard ΛCDM frame-

work, this flatness is a dynamically unstable equilibrium, which primordial inflation resolves by driving Ω_K towards zero. Furthermore, the accelerated expansion of the Universe is attributed to a cosmological constant (Λ), representing approximately 70% of the total energy density today, yet lacking a fundamental microscopic origin. The observational pressure has recently intensified: the H0 Distance Network Collaboration has established $H_0 = 73.50 \pm 0.81\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at $\sim$1% precision [7], yielding a 7.1σ tension with flat ΛCDM (Planck+SPT+ACT) and a 5.0σ tension with BBN+BAO (DESI DR2). This makes the status of the flatness constraint, and of the geometric origin of Ω_Λ, worth stating precisely — which is the purpose of this Letter. The vacuum energy catastrophe—a discrepancy of ≈ 55 (EW) to ≈ 122 (cutoff) orders between QFT predictions and the observed Λ [10, 11]—remains unresolved; the exact-flatness postulate adopted here does not resolve it either, and we make no claim that it does.

In this Letter, we introduce a paradigm shift regarding the nature of spatial flatness and cosmic acceleration. Extending general relativity to the Einstein-Cartan (EC) framework—where spacetime torsion is effectively algebraic in the low-density limit (ECKS [4, 9]), and acquires propagating degrees of freedom via Poincaré Gauge Theory (PGT) at high density [2]—we propose that spatial flatness is not an accident of initial conditions, but an *exact* property of the extended geometric framework — posited, not enforced by any dynamical response.

The key phenomenological calibration—the sound-horizon reduction and the effective amplitude that sets it, with the status of the S_8 sector under the internal closure $\Omega_m = \omega_m/h^2$—is presented in the companion paper Foundation I [1]; the microscopic dark sector and baryogenesis mechanism are developed in Foundation II [2]. This division of labor is deliberate: Foundation I provides the numerical calibration of (H_0, S_8, r_s), Foundation II supplies the microscopic torsion condensate and baryogenesis sectors, and the present Letter isolates what the Topological Invariance Principle does and does not underwrite — exact spatial flatness, and not the dark-energy sector.

THE POSTULATE, AND ITS EXACT SCOPE

What the postulate does *not* do

We state at the outset what this postulate cannot deliver, because earlier versions of this Letter claimed otherwise.

Once $k = 0$, dividing the Friedmann equation by H^2 gives $\sum_i \Omega_i = 1$ *identically*, for any matter content whatsoever: adding a spin component changes H, and the critical density $\rho_{\rm crit} = 3H^2/8\pi G$ follows. No deficit arises, hence nothing is compensated. The closure condition is an **identity**, not a constraint: it fixes neither $\rho_{\rm DE}$ nor its equation of state.

The point is sharpened by a comparison already made in Foundation II [2]. In pure ΛCDM, with ρ_Λ strictly constant ($w = -1$ exactly), Ω_Λ grows from 0.002 at $z = 10$ to 0.700 today, simply because $\rho_m \propto a^{-3}$ dilutes — an artefact of the denominator, not a strengthening of the dark sector. *A growing Ω is therefore not evidence of an evolving dark sector.* What would carry physical content is an evolving ρ, and the closure condition says nothing about that.

Relation to the elimination of k in unitary PGT. A nearby result deserves to be distinguished, because the two are easily conflated. Barker, Lasenby, Hobson and Handley (*Phys. Rev. D* **102**, 024048 (2020), arXiv:2003.02690) exhibit a ghost-and-tachyon-free Poincaré gauge theory in which *a conspiracy among the coupling constants eliminates the spatial curvature $k \in \{\pm 1, 0\}$ from the field equations.* That is not the present postulate, and the difference runs three ways. *(i) Content.* Here $\Omega_K \equiv 0$: space *is* flat. There, k may equal ± 1 and space may be curved — the dynamics simply does not see it. *(ii) Status.* Here the flatness is postulated as a topological necessity; there it follows from a condition on the couplings, that is, from a tuning rather than from a necessity. *(iii) Falsifiability, which is the decisive one.* A measurement of $\Omega_K \neq 0$ at any precision refutes the present postulate; it does *not* refute theirs, where k is inoperative rather than absent. *(iv) The two mechanisms are mutually exclusive.* This is worth stating plainly, because it removes any suspicion that the present postulate is a special case of theirs. Their k-screening requires *two* conditions, and the first is decisive: their Eq. (87) imposes a pure Yang–Mills theory, $\check{\alpha}_0 = 0$, which *removes the Einstein–Hilbert term* — "the traditional Einstein–Hilbert term is the first to be excluded". It is through that single coefficient that k enters their field equations, which is why setting it to zero screens the curvature. The framework used here retains $\tilde{R}$ linearly, with coefficient $1/2\kappa$, i.e. $\check{\alpha}_0 \neq 0$; and the same authors show, in their own Einstein–Cartan demonstration, that this case returns $\check{\alpha}_0 = \Omega_{\rm m} + \Omega_{\rm r} + \Omega_\Lambda + \Omega_K$, with Ω_K *present*. In an Einstein–Cartan theory the curvature is therefore not screened, by their own derivation: the mechanism is unavailable here, not by choice but by structure. *(v) A degeneracy that does not arise here.* One objection deserves a direct answer, since it would blunt the falsifiability just claimed. In Einstein–Cartan cosmology with totally antisymmetric torsion and *no* torsion potential, van de Venn *et al.* (*Eur. Phys. J. C* **83**, 288 (2023), arXiv:2211.11868) show that conservation forces the axial amplitude to dilute as a^{-1}, so that its energy density scales as a^{-2} with $w = -1/3$ — the equation of state of spatial curvature itself. The inferred geometry is then no longer fixed by k alone, and a

curved universe can be mistaken for a flat one; were that the situation here, a measurement of Ω_K would no longer test the postulate. It is not the situation here. The framework used in this work carries a torsion potential with a non-trivial vacuum expectation value, and it is the minimum of that potential — not conservation — that fixes the axial amplitude, pinning it to a constant. At the vacuum the condensate stress tensor is algebraic in the metric, so its density is independent of the scale factor and no a^{-2} piece appears: the sector is degenerate with Λ, not with curvature. The potential introduced for the dark sector thus protects the present postulate as a by-product. We state the scope plainly: this concerns the homogeneous background, and does not exclude a degeneracy arising from perturbations or from non-minimal torsion–curvature couplings. The reduction is reproducible symbolically via `Scripts_F2/verify_flrw_curvature_degeneracy.py`, which retains k throughout rather than imposing spatial flatness at the outset. The postulate stated here is therefore the more exposed of the two, and that is its point — but it is also the less derived, and we do not claim otherwise.

Withdrawn statements. Earlier versions of this Letter stated that any injected spin density algebraically enforces a compensating torsion debt, and that the torsion fraction increases mechanically to maintain topological equilibrium. Neither survives the argument above, and both are withdrawn. We also withdraw the analogy with charge conservation: in FRW, k is a constant of the solution, not a Noether charge — there is nothing to conserve.

What it does do

What remains is a sharp and *falsifiable* claim: spatial flatness is **exact**, $\Omega_K \equiv 0$, as a topological property rather than a dynamical outcome. This is not what inflation delivers: inflationary flattening predicts $\Omega_K \approx 0$, typically at the 10^{-5} level. A measurement of $\Omega_K \neq 0$ at *any* precision would exclude the present framework, while leaving inflation untouched. That asymmetry is the whole empirical content of the postulate — and, we argue, enough to make it worth stating.

We formulate the *Topological Invariance Principle* (TIP) as follows: *Global spatial flatness ($\Omega_{\rm tot} \equiv 1$) is an absolute and inviolable geometric equilibrium state, not a contingent initial condition. The same axial torsion field appears in two regimes across cosmic history—as a spin fluid at high density prior to recombination ($\Omega_{\rm spin}$), and as a frozen condensate at late times (Ω_τ)—whose scale-dependent structures are detailed in the companion papers [1, 2]. Whether a relation between the two follows from the action is the open problem posed in the closing section below; it is* not *supplied by the closure condition.*

Under this principle, the standard Friedmann equation for fractional densities is extended to include the spin-torsion pair:

$$\Omega_m + \Omega_r + \Omega_{\rm spin} + \Omega_\tau + \Omega_K = 1, \qquad \Omega_K \equiv 0 \ \ \forall\, t \quad (1)$$

where Ω_τ is the macroscopic torsion density that, in the standard parameterisation, plays the role of Ω_Λ: at leading order it acts as an effective cosmological constant, while its full dynamical equation of state $w_\tau(z)$ departs from -1 at late times, yielding $w_0 = -0.904$, $w_a = -0.153$ (calibration point, without error bar; the likelihood profile gives -0.907 ± 0.004 and -0.153 ± 0.007) [1].

The TIP raises $\Omega_K = 0$ from an observational coincidence to an exact topological property. It is *not* enforced dynamically by the pair ($\Omega_{\rm spin}$, Ω_τ): that these are two regimes of the same axial field is expected on physical grounds, but the relation between them is the open problem stated in the closing section, not a consequence of the postulate.

The companion papers use the identification

$$\Omega_\tau^{(\rm initial)} = -\Omega_{\rm spin}^{(\rm peak)} \quad (2)$$

This is an *empirical* identification, not a consequence of the closure condition (see the open problem below). The quantity $\Omega_{\rm spin}^{(\rm peak)} \approx 0.093$ is the spin-to-radiation ratio calibrated at the reference redshift $z_{\rm ref} = 5600$ in Foundation I [1]; the spin sector is expected to be sourced by the fermion spin density at the QCD transition ($T \sim 200$ MeV), but the evolution from that epoch to $z_{\rm ref}$ is not computed, and 0.093 is not a QCD-epoch value (with $\rho_{\rm spin}/\rho_r \propto (1+z)^2$, the same sector would give $\sim 10^{15}$ at the QCD epoch). It needs its a^{-6} law up to $z \sim 10^5$ and BBN excludes it at 1 MeV (Foundation I). *(Corrected 16/10/2026: earlier versions claimed the QCD physics fixes this value, without free parameter.)*

In the ultradense regime, the bounce corresponds to the right-hand side of the modified Friedmann equation

$$H^2 = \frac{8\pi G}{3}\big(\rho_r + \rho_m + \rho_\Lambda - \rho_{\rm spin}\big)$$

reaching zero at finite $a_{\rm min}$ — the non-singular bounce; no value of $\Omega_{\rm spin}^{(\rm peak)}$ is singled out by this condition together with $\Omega_{\rm tot} = 1$, which is an identity. The observational calibration of $\Omega_{\rm spin}^{(\rm peak)}$ from the H_0 and BAO sectors is described in Foundation I, where the S_8 branch of the calibration is historical (the growth role of the spin sector being open), [1], which obtains $\Omega_{\rm spin}^{(\rm peak)} \approx 0.093$.

Scope of the equation above. The Friedmann equation written here is the ultradense ECKS form, in which the unpolarised spin-spin contact term is repulsive and enters subtractively — the form that produces the bounce, and the one this section uses. The calibration quoted from Foundation I is obtained on a *distinct* effective background, in which the residual axial condensate is treated as a positive stiff component contributing $+\Omega_{\rm spin,0}(1+$

$z)^6$ to H^2. Its positive sign, stiff equation of state and isotropic reduction are calibrated properties of that branch, not consequences of the equation above; relating the two regimes is the same open problem already flagged for $(\Omega_{\rm spin}, \Omega_\tau)$ above. See Foundation I Extended, §*Sign convention*, for the full discussion.

Any future revision of that calibration would modify the numerical value entering Eq. (2), without altering the logical structure of the TIP itself. *Robustness note:* the TIP mechanism and its qualitative predictions (geometric dark energy, non-singular bounce, birefringence) hold for any value of $\Omega_{\rm spin}^{(\rm peak)}$ in a broad range, indicatively $[0.05, 0.15]$ (a range not derived from QCD thermodynamics). The precise value 0.093 is an observational calibration, not a logical prerequisite of the principle.

This identification is not derived. Whether an analogous relation holds continuously at all subsequent times is precisely the open question addressed in the closing section, not a constraint established here.

WHAT THE ACTION DOES GIVE, INDEPENDENTLY OF THE POSTULATE

Dark energy in this framework does *not* come from the closure condition. It comes from the action, and the derivation is independent of the postulate discussed above.

Reducing the axial torsion sector on an FLRW background (Foundation II [2], Appendix J; symbolically verified) gives three results. First, for $S_\mu = (s(t), 0, 0, 0)$ the field strength vanishes identically, $F_{\mu\nu}^{(S)} \equiv 0$: in a homogeneous universe this field *cannot* carry kinetic energy. Second, the spin source diluting as a^{-6}, the field settles at the minimum of its potential, $W'(X_0) = 0$. Third, at that minimum the stress tensor reduces to

$$T_{\mu\nu} = -\,W(X_0)\, g_{\mu\nu}, \tag{3}$$

exactly the form of a cosmological constant, whence

$$\rho = W(X_0), \qquad p = -W(X_0), \qquad w = -1. \tag{4}$$

The static part of dark energy is therefore **derived**, not posited — and it owes nothing to the postulate stated above.

Related results — and what is, and is not, new here. That a torsion contribution can behave as a cosmological constant is not a result of this framework alone: Ivanov and Wellenzohn obtained it by a different route, treating torsion as an external field [14], and a related $w \simeq -1$ behaviour arises in holographic dark-energy models with torsion [15], a mechanism of a different kind whose Hubble-cutoff variant is excluded in Einstein–Cartan cosmology [16]. The first of these indicates that the result is not an artefact of our particular reduction. *(Corrected 22/10/2026: formerly presented as independent derivations.) What is proper to the present framework is not the isolated result, but its integration* — the same axial torsion field supplies the spin fluid that reduces the sound horizon before recombination and the frozen condensate that sources Λ at late times. Whether that integration can be derived rather than observed is precisely the problem stated in the closing section.

The fluctuations give radiation, not dark energy. Writing $S_\mu = s_0\delta^0_\mu + \delta S_\mu$, the spatial components are massless Goldstone modes whose action is conformally invariant Maxwell; their mode equation $\delta S_k'' + k^2\delta S_k = 0$ carries no Hubble friction, so $\rho_{\delta S} \propto a^{-4}$ and $w = +1/3$. *The observed evolution of the equation of state does not follow from this sector either.* The pair (w_0, w_a) remains calibrated.

THE OPEN PROBLEM

We now state the question this Letter is written to pose.

The Weyssenhoff spin fluid that reduces the sound horizon before recombination (Foundation I [1]) and the frozen condensate that sources dark energy (previous section) are not two ingredients. They are **the same axial torsion field in two regimes**: dynamically sourced at high spin density, frozen at the minimum of its potential once that source has diluted. A relation between $\Omega_{\rm spin}$ and Ω_τ is therefore natural to expect on physical grounds.

What is *not* legitimate is to obtain it from Ω-bookkeeping. As shown above, the closure condition cannot supply such a relation: it is an identity. The link must come from the action, or not at all.

The question, stated precisely

Does the Poincaré Gauge action imply a relation between the spin density $\Omega_{\rm spin}$ that governs the pre-recombination sound horizon and the condensate density Ω_τ that sources late-time dark energy; and if so, with which sign?

We do not answer it here. We state it, show why the route through the closure condition fails, and indicate where a derivation might be sought.

Where a derivation might be sought. The obstacle is not a no-go theorem. The known no-go on crossing $w = -1$ for a single scalar field with Lagrangian $p(\phi, \partial\phi)$ [13] concerns a dynamical degree of freedom and does not directly cover the algebraically slaved torsion considered here; in any case it forbids a *crossing* which the data do not require: a non-crossing model already reproduces the calibrated (w_0, w_a) to within a few per cent. Moreover, the negative result of this section — that a diluting, algebraically slaved torsion does not carry late-time dark energy — has itself been reached independently, by a different route, by Izaurieta, Lepe

and Quinzacara [16]. What is needed is a torsion mode with a genuine kinetic term — a *propagating* axial mode; quadratic Riemann–Cartan terms already yield torsional dark-energy behaviour at the background level [12], and other groups are actively working along this line. Deriving its potential from the action, rather than positing it, is the outstanding task. Ghost-freedom and Hamiltonian stability would then have to be checked.

This does not resolve the open problem above, but it does fix the terminology used throughout the companion papers. The TIP mechanism operates in three stages: (i) a transient primordial spin density $\Omega_{\rm spin} \propto a^{-6}$, sourced by fermion spin at the QCD confinement transition ($T \sim 200$ MeV), is observationally calibrated at the acoustic-era reference redshift $z_{\rm ref} = 5600$, where it reaches its cosmologically relevant amplitude $\Omega_{\rm spin}^{(\rm peak)} \approx 0.093$ (Foundation I); (ii) as the spin fluid subsequently dilutes ($\rho_{\rm spin} \propto a^{-6} \to 0$), the axial field settles at the minimum of its potential and freezes into the vacuum structure, persisting as a residual geometric density that plays the observational role of Λ (previous section); (iii) the numerical relation $\Omega_\tau^{(\rm initial)} = -\Omega_{\rm spin}^{(\rm peak)}$ used in the companion papers is an *empirical* identification, not a consequence of the closure condition — establishing whether the action implies such a relation, and with which sign, remains the open problem stated above.

The numerical magnitude $\Omega_{\rm spin}^{(\rm peak)} \approx 0.093$ is calibrated observationally at $z_{\rm ref} = 5600$ in Foundation I [1] and is not computed from the QCD transition physics; it is reported here as an input, not re-derived. Crucially, while the kinetic spin fluid dilutes rapidly as a^{-6} due to cosmic expansion, the corresponding macroscopic torsion field freezes into the vacuum structure once the spin source vanishes.

Formally, within the PGT extension of the ECKS action detailed in Foundation II [2], this freezing corresponds to the axial torsion pseudo-vector $S_\mu \equiv \frac{1}{6}\varepsilon_{\mu\nu\lambda\rho}T^{\nu\lambda\rho}$ acquiring a non-zero vacuum expectation value,

$$\langle S_\mu \rangle = s_0\, \delta^0_\mu \neq 0, \tag{5}$$

whose energy density $\rho_\Lambda^{\rm eff} = \frac{1}{4}m_T^2 s_0^2$ — the depth of the tachyonic well — constitutes the effective cosmological constant at leading order. Its field-theoretic treatment, with the overall scale normalised on the observed dark-energy density, is developed in Foundation II [2]. At leading phenomenological order, this frozen torsion acts as an effective cosmological constant ($w_\tau = -1$). The calibrated late-time equation of state departs from this limit, with $w_0 = -0.904$, $w_a = -0.153$ (calibration point, produced by no deposited script; the likelihood profile of Foundation I gives -0.907 ± 0.004 and -0.153 ± 0.007; open problem PO-F1-W0 of Foundation I). *(Corrected 16/10/2026: earlier versions described this $w(z)$ as derived from the ECKS action and attached the ± 0.004 of the profile to -0.904.)* These values are calibrated, not derived: w_a is set through $\xi = 1.645$, a parameter calibrated on the DESI preference used by the ECF so as to reproduce the target value of w_a [1]; their consistency with the DESI BAO preference for $w \neq -1$ [5, 6] is therefore not a prediction independent of DESI. A full likelihood comparison is deferred to Foundation III. The values $w_0 = -0.904$ and $w_a = -0.153$ are therefore not free fit parameters of the Letter itself; they are inherited from the Foundation I calibration and serve here as the late-time phenomenological imprint of the same geometric torsion sector. As the Universe expands, the absolute densities of baryonic matter and topological dark matter [2] dilute as a^{-3}. Consequently, their fractional sum (Ω_m) decreases. Because the fractional densities sum to unity by construction once $k = 0$ (Eq. 1), the fractional weight of the frozen residual torsion (Ω_τ) rises over cosmic time as an arithmetical consequence of that dilution — the same rise a strict cosmological constant would show — not evidence of a new field or a compensating mechanism (see Figure 1).

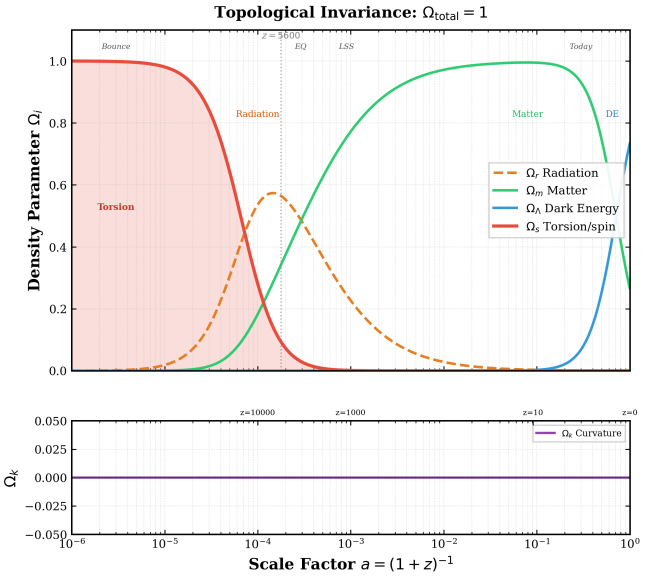


FIG. 1. **Topological Invariance.** Temporal evolution of fractional energy densities $\Omega_i(z)$ in the ECF model, computed with $F_{\rm ion} = 1.2765$, $\Omega_{\rm spin,peak} = 0.093$ [1]. The invariant $\Omega_{\rm tot} = 1$ (bottom line) is the visual expression of the TIP. The late-time rise of Ω_Λ is an arithmetical consequence of matter dilution — it occurs for a strict cosmological constant too — not the signature of a new field. Reproducible via `plot_Cosmic_History_Omegas.py`(public repo).

What the standard ΛCDM model interprets as a mysterious, dominant Dark Energy fluid ($\Omega_\Lambda \approx 0.7$) driving accelerated expansion is, in the ECF model, a frozen torsion condensate whose fractional weight grows as matter dilutes — an arithmetical consequence of that dilution, as for a strict cosmological constant, and not a response

that prevents the Universe from curving (the compensation reading is withdrawn, see the Conclusion).

CONCLUSION

It must be emphasised that the TIP operates here as a foundational *postulate* of an extended Einstein-Cartan framework, not as a theorem of the minimal ECKS Lagrangian. What Foundation II derives from the Poincaré Gauge Theory action is the torsion condensate that supplies a static dark-energy term via spontaneous symmetry breaking; the closure constraint $\Omega_{total} = 1$ itself remains a postulate, and its first-principles derivation—together with the $\Omega_{\rm spin}$–Ω_τ relation and its sign—is an open problem [2].

We close by separating what this Letter asserts, what it withdraws, and what it leaves open.

Asserted. Spatial flatness is exact, $\Omega_K \equiv 0$, by topological necessity rather than by dynamical relaxation. This is a falsifiable claim, and it is the whole content of the postulate: a measurement of $\Omega_K \neq 0$ at any precision would exclude the framework while leaving inflation untouched.

Withdrawn. The claim, present in earlier versions, that the closure condition enforces a compensating torsion debt and thereby generates dark energy. It does not: $\sum_i \Omega_i = 1$ is an identity once $k = 0$, and a growing Ω is equally a feature of a strict cosmological constant.

Left open. Whether the Poincaré Gauge action implies a relation between $\Omega_{\rm spin}$ and Ω_τ — the two regimes of a single axial torsion field — and with which sign. We regard this as the substantive question the framework now faces, and have stated it rather than assumed an answer. The static part of dark energy is *derived* from the FLRW reduction of the action, as shown above; its evolution remains calibrated, and relating it to the spin sector is the open problem stated in the section above. At leading order the frozen torsion mimics Λ; the calibrated $w(z)$ departs from it, in the direction DESI has since measured, though a full-covariance comparison places the point at 2.03σ from the DR1 central value and does not favour it over ΛCDM on those data alone (Foundation I). The detailed numerical work—the sound-horizon calibration of H_0 and the status of the S_8 sector (Foundation I [1]) and the modelling of topological dark matter via torsion knots (Foundation II [2])—is presented in the companion regular articles. A geometric consistency check of the frozen torsion debt yields a cosmic birefringence angle $\beta_{\rm ECF} \approx 0.35^\circ$ (Planck/WMAP baseline), 1.8σ from the ACT-DR6 result $0.215 \pm 0.074^\circ$ (itself a 2.9σ detection) [8] *(corrected 16/10/2026: earlier versions read the* 2.9σ *detection significance as a consistency level).* This is a calibrated consistency check, not a parameter-free prediction, and will be sharpened by LiteBIRD [1] (Sec. 7 of Foundation I).

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